Journal of Power Sources 438 (2019) 227063

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Role of borate functionalized carbon nanotube catalyst for the performance improvement of vanadium redox flow battery

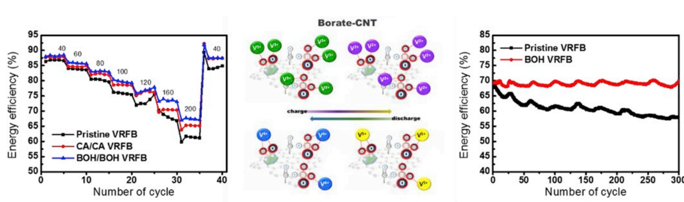
Yongjin Chung[a,1], Chanho Noh[b,1], Yongchai Kwon[b,*]

[a] Department of Chemical and Biological Engineering, Korea National University of Transportation, 50 Daehak-ro, Chungju, Chungbuk, 27469, Republic of Korea
[b] Graduate School of Energy and Environment, Seoul National University of Science and Technology 232 Gongneung-ro, Nowon-gu, Seoul, 01811, Republic of Korea

HIGHLIGHTS

- Borate-CNT is suggested as a catalyst for both side of VRFB.
- The performance and long term durability of VRFB is enhanced by Borate-CNT.
- Co-treatment by NaOH and boric acid increase the active site on the surface of CNT.
- Boron-oxygen bonds promote the redox reaction of vanadium ions.
- VE and EE are improved considerably by adoption of Borate-CNT on carbon felt.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Vanadium redox flow battery
Borate group functionalized carbon nanotube
Boron-oxygen bonds
Electronegativity
Co-treatment of NaOH and H_3BO_3

ABSTRACT

A borate-group functionalized carbon nanotube (Borate-CNT) catalyst is developed to enhance the performance and long term durability of the vanadium redox flow battery (VRFB). To prepare for the –Borate-CNT catalyst, carboxylic acid groups (COOHs) adopted onto acidified CNT (CA-CNT) are simply co-treated with sodium hydroxide (NaOH) and boric acid (H_3BO_3). As a result, the COOHs are transformed into borate groups. The transformation is verified by XPS analysis, showing the increase in oxygen content and the creation of boron-oxygen bonds. The active sites and catalytic activity of Borate-CNT are increased more than those of the catalysts formed by the single treatment of NaOH and CA-NCT or H_3BO_3 and CA-CNT. This is due to the increase of active sites by the formation of oxygen abundant borate groups and the different electronegativity between the boron and oxygen elements promotes the attraction and subsequent reaction o f vanadium ions. The voltage and energy efficiencies (VE and EE) of the VRFB using Borate-CNT catalyst are better than those of VRFBs using no catalyst or CA-CNT catalyst – even at 200 mA cm^{-2} – and the efficiencies of Borate-CNT VRFB are well maintained until 300 cycles, whereas the efficiencies of the no catalyst VRFB are considerably decreased (17% and 16% decreases of the initial values in VE and EE). In addition, Borate-CNT shows the effect of protection by suppressing the chemical aging of carbon felt from toxic acidic electrolyte.

1. Introduction

Due to the increasing concerns over fossil fuels causing various problems – such as pollution, global warming, and an unstable supply – the need for development and actual application of sustainable energy has increased. Therefore, effort for establishing proper renewable

* Corresponding author.
E-mail address: kwony@seoultech.ac.kr (Y. Kwon).
[1] Y. Chung and C. Noh contributed equally to this work.

https://doi.org/10.1016/j.jpowsour.2019.227063
Received 31 March 2019; Received in revised form 6 August 2019; Accepted 22 August 2019
Available online 24 August 2019
0378-7753/

energy systems including wind, solar, hydropower, and geothermal has been made. However, in spite of such efforts, proper renewable energy systems have not been developed yet because they cannot generate electricity regularly. Thus, the widespread application of renewable energy systems should be accompanied by large-scale energy storage system (ESS) because the ESS can compensate for the unbalance occurring between electricity supply and demand, and assist in the stable operation of renewable energy systems [1–3].

Although many researchers have demonstrated various types of ESSs – such as the pumping-up power station, flywheel, hydraulic, thermal accumulator, etc. – electrochemical-based devices, also known as rechargeable batteries, have been the most commonly considered because of the freedom from geographical dependency and the utilization of relatively small volume [4,5]. Of them, the redox flow battery (RFB) has received the spotlight as the proper solution for ESS due to high energy efficiency, independent utilization of power and capacity, short response time, and long cycle life [6,7].

RFB is mainly classified by its active species and transition metal ions such as vanadium, ferric, zinc ions, and some organic compounds were mostly used due to their outstanding characteristics associated with redox reactivity. All vanadium RFB (VRFB), which utilizes the differently oxidized vanadium ions for both the anolyte and catholyte, has been widely considered because it can alleviate the issue of cross-contamination by using the same vanadium ions as both electrolytes and induce excellent reaction reversibility [8,9]. Many researchers have made efforts to further enhance the performance of the VRFB via increasing the slow redox reaction rates of electrodes. For this purpose, the adoption of new catalysts is important and, to date, carbon-based materials like graphitic nanostructures have been preferred because of the relatively low price and easy preparation, excellent electrical conductivity, and applicable long term stability in harsh electrolyte pH conditions. Additionally, the effects of adoption on the performance of VRFB was compared with the effects of other candidates like noble metals and metal oxides [10–16].

In detail, doping oxygen-based functional groups onto the surface of carbon materials is a well-known strategy to enhance the redox reactivity of vanadium ions because the oxygen functional groups can become the active sites for the redox reaction of vanadium ions and increase the hydrophilicity of the electrode, which can promote the reactivity of vanadium ions dissolved in the water-based electrolyte [17–21]. Hence, most of the studies about the carbon materials applied as electrodes for VRFB have been focused on increasing the content of oxygen functional groups by thermal and chemical treatments. However, excessive oxygen functional groups may lead to a negative effect on long term stability by the degradation and weakness in mechanical strength of the carbon materials, so it is almost mandatory to find new ways to improve the redox reactivity of vanadium ions without a large impact on the mechanical properties [20,22,23].

As a meaningful alternative, in this paper, a borate-group functionalized carbon nanotube (CNT) and its easy preparation procedure are suggested as the catalyst to not only maintain long term stability but to also enhance the redox reaction of vanadium ions in both electrodes. To do this, carboxylic acid groups are functionalized onto CNT (CA-CNT) and the CA-CNT is treated by an easy wet process through both sodium hydroxide (NaOH) and boric acid (H_3BO_3). As a result, the oxygen content of the catalyst was significantly increased by the formation of abundant borate functional groups, and its catalytic activity for the redox reaction of vanadium ions was improved, while maintaining the durability of the VRFB. To evaluate the effects of NaOH and H_3BO_3 treatments on the chemical structure of CA-CNT, X-ray photoelectron spectroscopy (XPS) and electrochemical evaluation are performed and VRFB single cell tests are conducted to examine whether or not the new catalyst is effective to increase the performance of VRFB and to compare its effect with that obtained by the other widely considered candidates.

2. Experimental

2.1. Materials

Carboxylic acid functionalized multi-walled carbon nanotube (CA-CNT) [thin, extent of labelling: >8% carboxylic acid functionalized, avg. diam. × L 9.5 nm × 1.5 μm], boric acid (H_3BO_3) (BioReagent, for molecular biology, suitable for cell culture, suitable for plant cell culture, ≥99.5%), Vanadium (IV) oxide sulfate hydrate (97%), and Nafion 117 solution (~5% in a mixture of lower aliphatic alcohols and water) are purchased from Sigma-Aldrich and used. A 3 mm thick PAN graphite felt (XF30A, Toyobo, Japan) is used. Sodium hydroxide (NaOH) (bead, >98% Samchun chemical, Korea) and Sulfuric acid (H_2SO_4) (95.0%, Samchun chemical, Korea) are also used.

2.2. Preparation of NaOH/CA-CNT and BA-CNT

For the NaOH treatment, 40 mg of CA-CNT was mixed with 10 mL of the 1 M NaOH solution for 48 h to improve the catalytic activity of CA-CNT for the VO^{2+}/VO_2^+ redox reaction. For preparing NaOH and H_3BO_3 co-treated CNT (BA-CNT), 40 mg of CA-CNT was mixed with 10 mL of the boric acid solution (3.65, 7, and 11 pH) for 48 h. The boric acid solution was made by first mixing 5.7 g of boric acid powder with 100 mL of deionized water (DIW) while stirring, and then changing the pH by adding 1 M NaOH solution. Finally, each mixture solution was centrifuged and purified with DIW at least 3 times to remove unreacted reactants and impurities, and then dried at 60 °C for 12 h.

2.3. Catalytic activity and reversibility measurements

To evaluate the electrochemical properties of each catalyst for the vanadium ions redox reaction, a VSP-128 potentiostat (Bio-Logic) was employed for cyclic voltammogram (CV). Cyclic voltammetry was carried out in a 3-electrochemical cell with a Pt wire as the counter electrode, Ag/AgCl as the reference electrode, and glassy carbon electrode (GCE, 0.1963 cm^{-2}) as the working electrode. The corresponding catalyst was coated on the GCE. A quantity of 4 mg of the catalyst powder was dissolved in 400 μL DIW under sonication for preparation of 20 μL of each catalyst to be loaded on the GCE. After the catalyst ink was dried, 5 μl of Nafion 117 solution diluted 1/10 with isopropanol alcohol (IPA) was loaded to fix the catalyst to the electrode surface and to block the pores within electrode surface. With the blockage of pores, semi-infinite diffusion condition can be adopted to interpreting the redox reaction of vanadium ions [24]. A solution consisting of 2 mL of 1.5 M $VOSO_4$ that was dissolved into 2 mL of 2 M H_2SO_4 solution and 18 mL of H_2O was used as the electrolyte [25–27].

2.4. Fabrication of catalysts coated felts

For investigating the effect on VRFB performance of the catalyst coated on the graphite felt, the catalysts were coated onto the surface of the graphite felt. First, 20 mg of catalyst powder was mixed with 7 mL of IPA, followed by ultrasonically dispersing for 30 s. Then, graphite felt was submerged in each catalytic ink solution and shaken for 10 min. Next, each felt was mixed with 19.9 mL of IPA and 0.1 mL of Nafion 117 solution and shaken for 5 min to coat the catalyst on the graphite felt surface. Finally, the catalyst coated graphite felt was dried at 60 °C for 12 h. The VRFB single cell tests sequentially underwent a pre-activation step, a charging step, and a discharging step. Regarding the detailed process, the same procedure that was previously reported was used [27, 28].

2.5. VRFB single cell fabrication and cyclic tests of charge-discharge sequence

The VRFB single cells were configured similarly to previously reported ones [27–29]. As both the positive and negative electrodes, graphite felts were prepared. The total active area of the VRFB single cell is 4 cm^2 and Nafion 117 is used as the membrane. The tanks connected with both negative and positive electrodes are filled with a solution containing 1.5 M $VOSO_4$ + 2 M H_2SO_4. The $VOSO_4$ and H_2SO_4 solutions are mixed under vigorous stirring until a clear blue color is observed, indicating VO^{2+}. The vanadium solution was utilized in both the positive and negative tank at 17 mL and 15 mL, respectively. To avoid overcharging, there is a volume difference between the positive and negative tanks. At first, both tanks use the VO^{2+} solution to charge the voltage to 1.7 V. After the activation step, the positive tank is filled with the solution containing VO^{2+}, while the negative tank is filled with the solution containing the V^{3+} solution. The range of cut-off voltage is from 1.7 V (for charging) to 0.8 V (for discharging). To examine how current density affects the stability of the VRFB, the current density used for the operation of the VRFB single cell was from 40 to 200 mA cm^{-2}. In addition, to evaluate the long-term stability of the VRFB single cell, the cell is charged and discharged at 200 mA cm^{-2} for 300 cycles. All the VRFB single cell tests were conducted under fixed temperature (25 °C) and flow rate (16.8 mL min^{-1}).

3. Result and discussion

3.1. The effect of NaOH and H_3BO_3 treatment on chemical structure of catalyst

To evaluate the bond structure and atomic concentration of the prepared catalysts, XPS analysis was performed and the result is represented in Fig. 1. The catalysts were prepared for investigating the effects of NaOH and/or H_3BO_3 adopted on CNT or CA-CNT. According to Fig. 1, the O1s ratios of Borate-CNTs with pH of 3.65, 7, and 11 (Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, and Borate-CNT @ pH 11) were 4.9%, 10.9%, and 22.7% respectively, while CA-CNT and NaOH/CA-CNT were at 5.6% and 9.9%, respectively.

"There are two significant findings about this result. First, oxygen content was increased by NaOH treatment of CA-CNT. This is the opposite to previous results reporting that the NaOH treatment of CA-CNT caused the decrease in oxygen content due to the removal of amorphous carbon and oxidation debris preventing a proper functionalization of the CA-CNT [30]. However, the other report suggested amorphous carbon like carboxylated carbonaceous fragments (CCFs) can converted to metal (sodium) carbonates (CO_Xs) by the reaction between NaOH and CCF, and the surface area density and oxygen atom contents can be increased considerably by the partial opening of CNT tips by NaOH treatment [31]. This means that the NaOH treatment can affect the surface property of the CA-CNT by three different ways: the removal of the CCF impurities from CNT surface, increases in oxygen content and surface area density by the ring opening of the CNT tip, and CO_Xs generation from the defected CNTs. Second, as the pH of the solution containing H_3BO_3 increased (NaOH concentration increased), the oxygen content of the Borate-CNT also increased. This means the NaOH and Borate ion make a synergetic effect to form oxygen abundant group. In other words, borate ions form in high pH condition and the carboxylic acid and hydroxyl groups observed on the sidewall of the CA-CNT can produce the oxygen atom abundant borate (BO_X) groups by the dehydration condensation reaction or physical interaction via hydrogen bonds formed between the borate ions and carboxylic acid groups of CA-CNT [32,33], meanwhile the carbonate formed from the CCF can be converted into tetraborate (B_yO_z) as shown in Fig. 2 [31]. This implies that, regardless of the type of carbon materials, the oxygen content of CNT-based catalysts increases considerably in high pH conditions, whereas this is not valid in low pH conditions because the carbonate groups may not be created from the CCFs. Moreover, in the Borate-CNT @ pH 3.65 catalyst, the borate groups may not properly attach to the sidewall of the CNT due to the interruption of the amorphous carbon and oxidation debris. Analysis of Na1s peak density also supported the previous explanation. In 1 M NaOH, the Na1s (1070 eV) to C1s (285 eV) ratio of the NaOH/CA-CNT catalyst was only 4.43 to 85.71 (5.2%), while that of the Borate-CNT @ pH 11 was 9.9–67.36 (14.7%). This implies that more sodium elements are captured in the Borate-CNT catalyst due to the strong chelation capability of the borate group even in a low sodium concentration.

To evaluate above assumption, the analysis of the O1s peak was conducted to investigate the difference in chemical composition of each catalyst, (Fig. 3). In the CA-CNT catalyst, only two main peaks representing organic-O bonds (C–O, 532 and C═O, 533 eV) were observed (Fig. 3a), while in the NaOH/CA-CNT catalyst, two more obvious peaks showing Na KLL auger (536 eV) and a metal-O bond (531 eV) were observed (Fig. 3b). Here, the peak density of metal-O bonds was obviously higher than that of the organic-O bonds."

To further investigate the difference in chemical composition of each catalyst and evaluate above assumption, the analysis of the O1s peak was conducted (Fig. 3). In the CA-CNT catalyst, only two main peaks representing organic-O bonds (C–O, 532 and C═O, 533 eV) were observed (Fig. 3a), while in the NaOH/CA-CNT catalyst, two more obvious peaks showing Na KLL auger (536 eV) and a metal-O bond (531 eV) were observed (Fig. 3b). Here, the peak density of metal-O bonds was obviously higher than that of the organic-O bonds. From Figs. 1 and 3, it can be speculated that the main oxygen bonds in NaOH/CA-CNT turns into metal-O bonds as the oxygen concentration of the catalyst by NaOH treatment increases, while that of CA-CNT turns into organic-O bonds. In other words, with the proper adoption of functional groups and the increase in amount of oxygen atoms, the substrate can interact well with metal ions due to the formation of ligands that contain oxygen atoms.

As the pH for the fabrication of different Borate-CNT catalysts increases from 3 to 11 (Fig. 3c–e), the XPS peak shape turned from the type of CA-CNT to that of NaOH/CA-CNT. At a pH of 11, H_3BO_3 turns into the BO_X ion and the ratios of metal-O bonds to Na KLL augur and C–O to C═O became higher than those of NaOH/CA-CNT catalyst, even though the amount of NaOH (or OH- ion) in the Borate-CNT catalyst was considerably low. The increase in the ratio of metal ion bonds to Na KLL augur implies that in the Borate-CNT catalyst, more oxygen groups participate in the interaction with sodium ions, and this indicates that the oxygen structure of NaOH/CA-CNT is different from that of Borate-CNT @ pH 11. In addition, the increase of the C–O bond which represents the formation of the B–O bond is clear evidence that the BO_X-based functional groups are well attached to the CNTs and they can interact with metal ions in a better way.

Altogether, NaOH and H_3BO_3 co-treatment increases the amount of oxygen atoms in CNT and BO_X-based functional groups, and they may

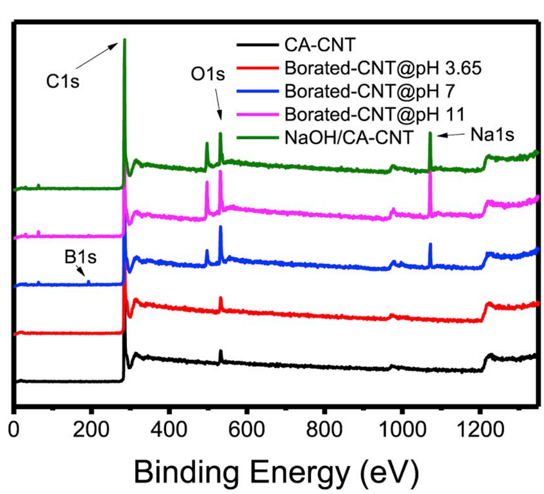


Fig. 1. The overall XPS analysis results of CA-CNT, Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, Borate-CNT @ pH 11, and NaOH/CA-CNT.

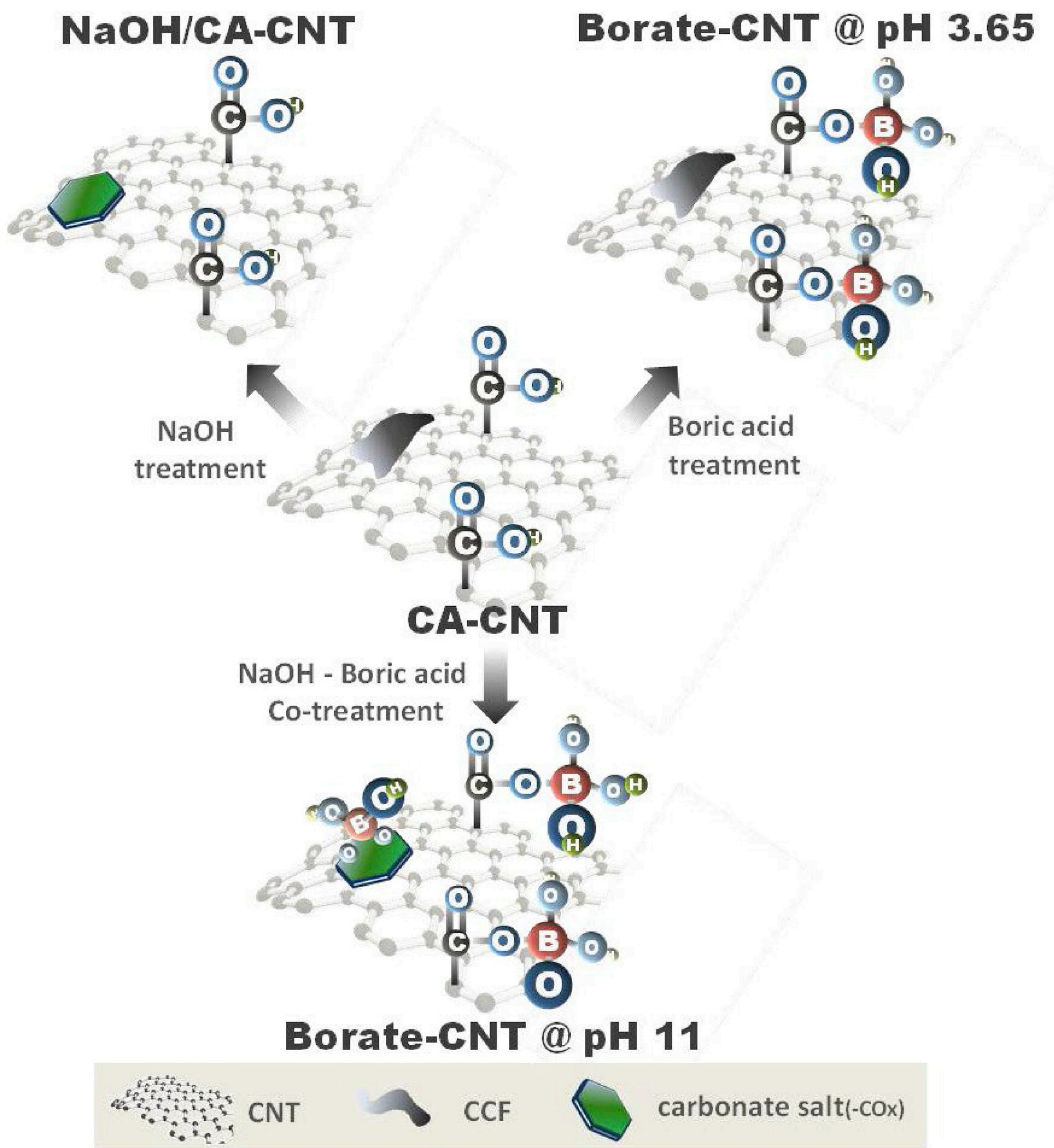


Fig. 2. A schematic showing the synthetic procedure of NaOH/CA-CNT and Borate-CNT.

form a strong and proper interaction with metal ions, such as vanadium ions, which would then affect the performance of VRFB, considerably.

3.2. *Redox reactivity of vanadium ions by the corresponding catalysts*

Based on the XPS analysis, it is confirmed that the oxygen content of CA-CNT can increase via the NaOH and H_3BO_3 treatments. However, the effect of the Borate-CNT catalyst on the redox reaction with vanadium ions cannot easily be predicted because of the differences in chemical structure of the catalyst as already explained. For example, the NaOH treatment of CA-CNT induces the separation of CCF and CNT. Although it can be a merit in terms of the smooth electron transfer via the tunneling effect between the sidewall of the CNT (active site) to the CNT, the treatment can also remove the active sites of the CNT surface [20]. Moreover, although it is expected that the increase in the amount of oxygen atoms will enhance the catalytic activity of the catalyst, it may also act as a barrier for electron transfer if the oxygen atoms form excessively on the CNT surface [34].

To verify the effects of the NaOH and H_3BO_3 treatments on the redox reactions of the vanadium ions, CV curves were measured and the results are summarized in Fig. 4 and Tables 1 and 2. According to Fig. 4, when the anodic peak current density (J_{pa}) of CA-CNT, NaOH/CA-CNT, and three Borate-CNT catalysts (Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, and Borate-CNT @ pH 11) were measured, the values were 3.5, 4.1, 4.0, 5.6, and 6.5 mA cm^{-2}, respectively, and even the cathodic peak current density showed a similar tendency as the anodic peak current density. More specifically, regarding the redox reaction rate of VO^{2+}/VO_2^+, the peak current densities of the NaOH/CA-CNT and the Borate-CNT @ pH 3.65 catalysts were 17.2% and 12.4% more improved than that of the CA-CNT catalyst, while the Borate-CNT @ pH 11 catalyst co-treated with NaOH and H_3BO_3 surprisingly showed an enhancement of 83.0%. This catalytic activity result supports the trend of the chemical bonding analysis (Fig. 1), showing the positive effect of the NaOH and H_3BO_3 co-treatment.

The co-treatment also promoted the formation of oxygen rich functional groups and improved the catalytic activity of the catalysts on VO^{2+}/VO_2^+ redox reaction. However, for the V^{2+}/V^{3+} redox reaction, the previously mentioned catalysts showed a J_{pa} of 3.5, 3.2, 3.6, 4.2, and

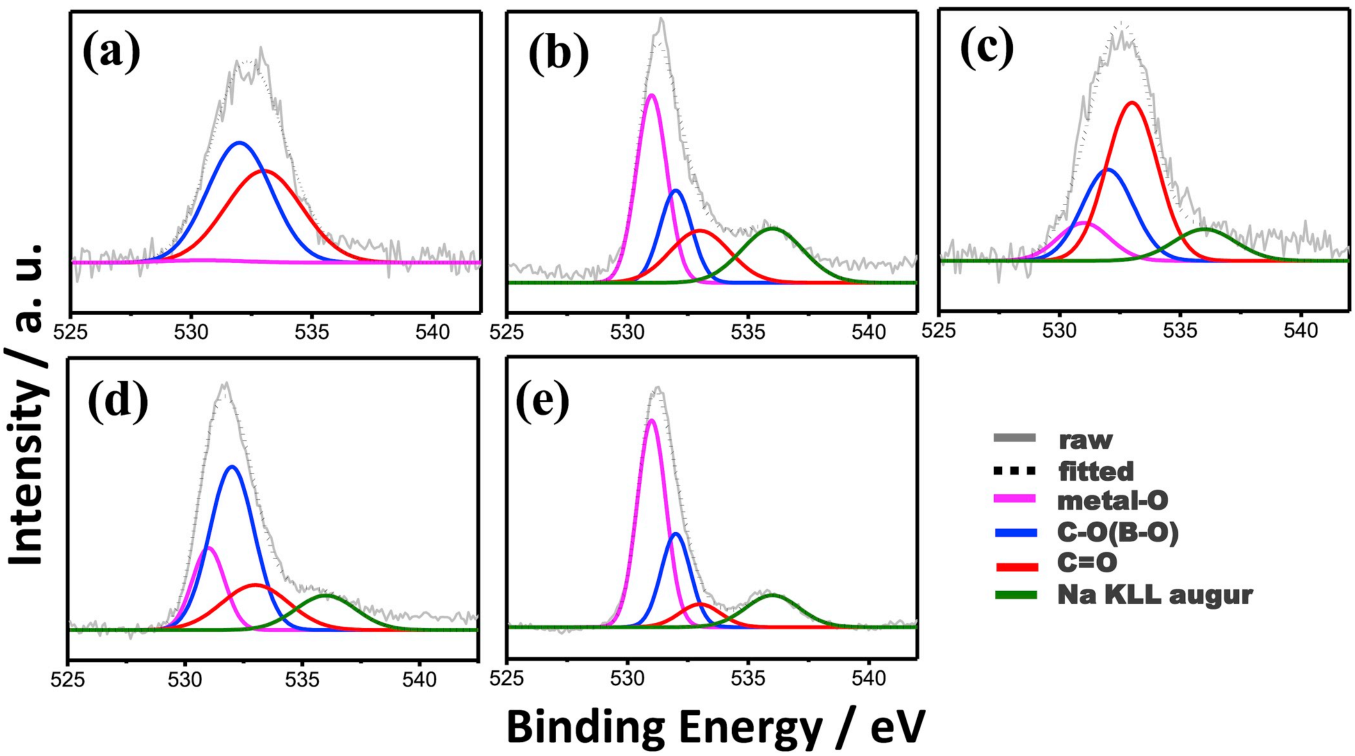


Fig. 3. The O1s analysis results of (a) CA-CNT, (b) NaOH/CA-CNT, (c) Borate-CNT @ pH 3.65, (d) Borate-CNT @ pH 7, and (e) Borate-CNT @ pH 11.

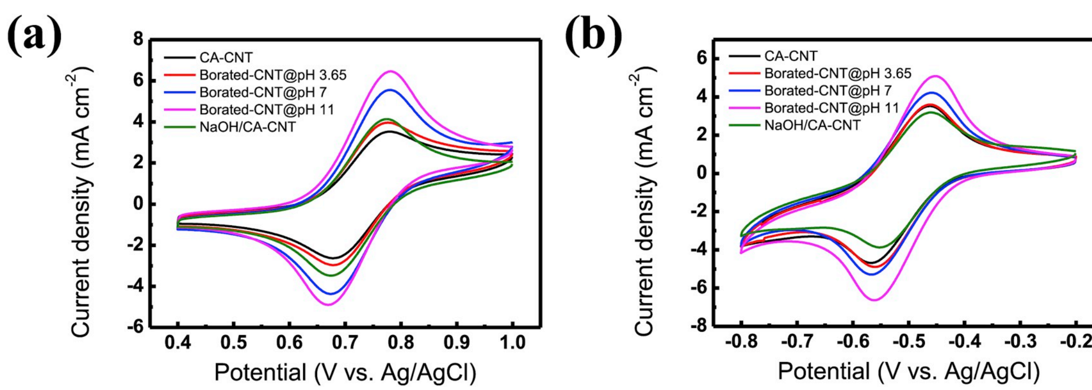


Fig. 4. CV curves of CA-CNT, NaOH/CA-CNT, Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, and Borate-CNT @ pH 11 representing (a) VO^{2+}/VO_2^+ redox reaction measured at the electrolyte consisting of 0.15 M $VOSO_4^+$ + 0.2 M H_2SO_4, and (b) V^{2+}/V^{3+} redox reaction measured at the electrolyte consisting of 0.15 M V^{3+} + 0.2 M H_2SO_4. For the tests, scan rate of 10 mv s^{-1} was used.

Table 1
Peak current density, peak potential and difference in peak potential and redox reaction reversibility for both VO^{2+}/VO_2^+ redox reactions of CA-CNT, NaOH/CA-CNT, Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, and Borate-CNT @ pH 11. For the tests, scan rate of 10 mv s^{-1} was used.

VO^{2+}/VO_2^+		Peak current density (mA cm^{-2})		Peak potential (V)		ΔE_p (V)	Ratio (I_{pc}/I_{pa})
		Anodic	Cathodic	Anodic	Cathodic		
CA-CNT		3.5	−2.6	0.78	0.68	0.1	0.75
NaOH/CA-CNT		4.1	−3.5	0.78	0.67	0.11	0.84
Borated-CNT	@pH 3.65	4.0	−3.0	0.78	0.68	0.1	0.75
	@ pH 7	5.6	−4.4	0.78	0.68	0.1	0.79
	@ pH 11	6.5	−4.9	0.78	0.67	0.11	0.76

5.1 mA cm^{-2}, respectively, meaning that for the V^{2+}/V^{3+} redox reaction, the single treatment was not effective for increasing the catalytic activity. Specifically, a single treatment of NaOH or H_3BO_3 just increases the amount of oxygen atoms, but co-treatment induces the remarkable enhancement even in catalytic activity as well as the increase in the amount of oxygen atoms.

To further investigate the effects of the differently treated catalysts on the redox reaction of vanadium ions, the redox reaction reversibility was measured (Table 1). According to the result, the reaction reversibility of the NaOH/CA-CNT and Borate-CNT @ pH 11 catalysts was

Table 2
Peak current density, peak potential and difference in peak potential and redox reaction reversibility for both V^{2+}/V^{3+} redox reactions of CA-CNT, NaOH/CA-CNT, Borate-CNT @ pH 3.65, Borate-CNT @ pH 7, and Borate-CNT @ pH 11. For the tests, scan rate of 10 mv s^{-1} was used.

V^{2+}/V^{3+}		Peak current density (mA cm^{-2})		Peak potential (V)		ΔE_p (V)	Ratio (I_{pc}/I_{pa})
		Anodic	Cathodic	Anodic	Cathodic		
CA-CNT		3.5	−4.7	−0.46	−0.57	0.107	1.329
NaOH/CA-CNT		3.2	−3.9	−0.46	−0.56	0.093	1.215
Borated-CNT	@ pH 3.65	3.6	−4.9	−0.46	−0.56	0.100	1.360
	@ pH 7	4.2	−5.3	−0.46	−0.57	0.108	1.253
	@ pH 11	5.1	−6.6	−0.45	−0.56	0.109	1.304

better than that of the CA-CNT catalyst, while BA-CNT @ pH 3.65 was affected in a negative way. This implies that the NaOH treatment is crucial for improving the redox reaction reversibility of vanadium ions, and the removal of impurities from the CNT surface is the key factor for enhancing the redox reaction reversibility of vanadium ions.

To determine the rate determining step, the kinetic parameters representing the redox reactivity of the vanadium ions were calculated (Fig. 5). According to Fig. 5, all the current densities were proportional to the square root of the scan rate, meaning that the redox reaction of the vanadium ion is controlled by mass transfer, not by the surface reaction. In turn, the active area (*A*) of the vanadium ions in the corresponding catalysts was calculated by using the Randle-Sevcik equation ($I_p = 269n^{3/2}AD^{1/2}\vartheta^{1/2}C$) and the result is summarized in Table 3. In the Randle-Sevcik equation, n is the number of electrons, A (cm^2) is the active area of catalyst, D ($cm^2\ s^{-1}$) is the diffusion coefficient, ϑ (V s^{-1}) is the scan rate and C (mol L^{-1}) is the bulk concentration of the solution [28,35].

As shown in Table 3, the $AD^{1/2}$ values of CA-CNT, NaOH/CA-CNT and Borate-CNT @ pH 3.65, Borate-CNT @ 7, and Borate-CNT @ 11 catalysts were 2.13×10^{-4}, 2.58×10^{-4}, 2.53×10^{-4}, 3.15×10^{-4}, and $3.74 \times 10^{-4}\ cm^3\ s^{-1/2}$, respectively. Here, the Borate-CNT @ 11 catalyst has the highest $AD^{1/2}$, meaning that its active area (A) is the highest because the diffusion coefficient of the vanadium active species (D) is supposed to be constant regardless of the catalyst. Thus, this result supports previous results and confirms that the Borate-CNT @ 11 catalyst fabricated with the NaOH and H_3BO_3 co-treatment can induce an increase in both the active area and reaction rate of vanadium.

3.3. Mechanism about redox reaction of vanadium ions

By XPS analysis and CV measurement, the catalytic activity of the borate functionalized CNT for the redox reactivity of vanadium ions can be explained as follows: in many previous researches publishing the effect of catalysts in VRFB, the oxygen functional groups are the most important for promoting the redox reactivity of vanadium ions [18,19,23]. Namely, the negatively charged oxygen functional groups, such as

Table 3
The $AD^{1/2}$ value of the CA-CNT, NaOH/CA-CNT, Borate-CNT @ pH 3.65, Borate-CNT @ 7, and Borate-CNT @ 11 catalysts calculated by the Randle-Sevcik equation.

	CA-CNT	NaOH/CA-CNT	Borated-CNT		
			@pH 3.65	@pH 7	@pH 11
$AD^{1/2}$ ($cm^3\ s^{-1/2}$)	2.13×10^{-4}	2.58×10^{-4}	2.53×10^{-4}	3.15×10^{-4}	3.74×10^{-4}

the hydroxyl and carboxylic acid groups, attract the positively charged vanadium ions by electrostatic attraction. In turn, their interaction plays a critical role in forming the chelated intermediates of the vanadium ions and lone electron pairs in oxygen and then further changes the oxidation state of the vanadium ions by exchanging electrons with the electrodes [27]. This indicates that the redox reactivity of the vanadium ions is dependent on the amount and polarity of functionalized oxygen groups.

In this regard, the borate functionalized CNT catalyst co-treated with NaOH and H_3BO_3 had a higher oxygen concentration than other catalysts (Fig. 1), and its catalytic activity was also better than that of other catalysts (Fig. 4). Regarding the polarity of oxygen, that of the oxygen atom belonging to the B–O bonds within Borate-CNT is higher than that of the C–O bonds belonging to the acidified carbon structure of CA-CNT. This is due to a higher electronegativity difference of the boron and oxygen elements than that of the carbon and oxygen elements, and this leads to (i) an increase in the polarity of the oxygen atom belonging to the B–O bond and (ii) a proper interaction between the oxygen atom and vanadium ion.

In terms of oxygen configuration, for the VO^{2+}/VO_2^+ redox reaction, the borate group is more appropriate than the oxygen group of hydroxyl and carboxylic acids. It is known that in the VO^{2+}/VO_2^+ redox reaction, two adjacent oxygen atoms are needed to interact with one vanadium ion via the process step of adsorption and surface reaction [28,36]. In the case of the oxygen group within the hydroxyl and carboxylic acid,

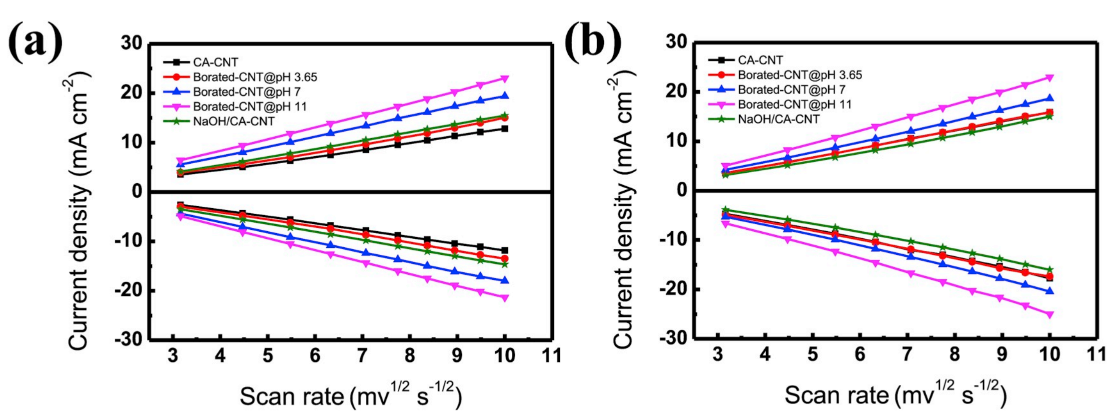


Fig. 5. Relationships between the square root of the scan rate and (a) the VO^{2+}/VO_2^+ peak current density and between the square root of the scan rate and (b) the V^{2+}/V^{3+} peak current density.

only one oxygen functional group can have contact with the vanadium ion, thus, the VO^{2+}/VO_2^+ redox reaction rate is low even though the overall amount of oxygen atoms is sufficient [27,28]. However, as shown in Fig. 4a, Borate-CNT has a different oxygen configuration, meaning that the Borate-CNT has two adjacent oxygen atoms that can interact with one vanadium ion, which can facilitate the VO^{2+}/VO_2^+ redox reaction and this is confirmed by the electrochemical characterization.

In summary, Borate-CNT can induce a better redox reactivity of vanadium ions due to the formation of oxygen abundant borate groups, a high polarity, and a proper configuration of oxygen atoms. The schematic of the mechanism is represented in Fig. 6.

3.4. VRFB single cell tests using the corresponding catalysts

To elucidate the effect of the borate functional group on the performance and durability of the VRFB single cell, VRFB single cells using pristine carbon felt and catalyst covered carbon felts (CA-CNT and Borate-CNT) were prepared and their performance was evaluated. They are denoted as Pristine VRFB, CA-CNT VRFB, and Borate-CNT VRFB, respectively. In all of the VRFB single cells, each carbon felt was used as both electrodes.

To investigate the performance of the three VRFB single cells under various current densities, their efficiencies were measured (Fig. 7). For doing that, the efficiencies were initially measured at 40 mA cm^{-2}, and then the current density was gradually increased up to 200 mA cm^{-2} by

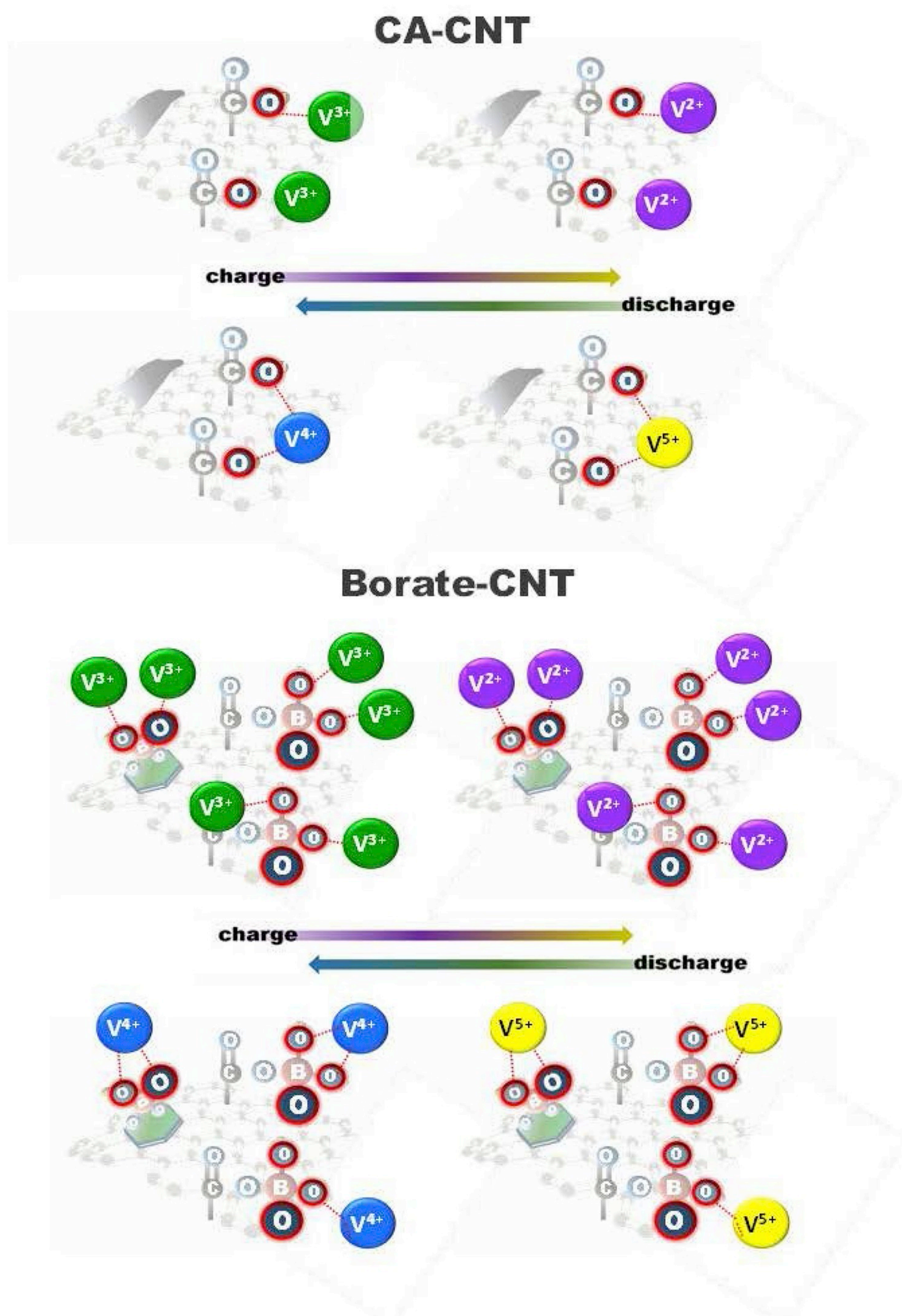


Fig. 6. The suggested redox reaction mechanisms of vanadium ions with CA-CNT and Borate-CNT catalysts.

increments of 20 mA cm^{-2} while the efficiencies of the VRFBs were continuously measured. For each current density, data was recorded for five cycles. According to Fig. 6a, the charge efficiency (CE) of the CA-CNT VRFB and Borate-CNT VRFB was 96% while that of the Pristine VRFB was 94%. The difference in CE is probably due to the inner pressure of the VRFB single cells increasing from the adoption of the catalyst and the volume decrease of the electrolytes. Due to this, the crossover of vanadium ions decreases in the CA-CNT VRFB and Borate-CNT VRFB [28,37].

Next, voltage efficiency (VE) of the three VRFB single cells was measured (Fig. 7b). As shown in Fig. 7b, at low current densities (40 and 60 mA cm^{-2}), the VE of all of the VRFB single cells was similar (88%), but after 80 mA cm^{-2}, the Borate-CNT VRFB maintained its VE almost entirely, while VE of the CA-CNT VRFB and Pristine VRFB reduced gradually. More specifically, Pristine VRFB, CA-CNT VRFB, and Borate-CNT VRFB at 200 mA cm^{-2} were 64%, 68%, and 71%, respectively, whereas at 80 mA cm^{-2}, they were 85%, 86%, and 87%, respectively. This means that when the current density is high (200 mA cm^{-2}), in terms of VE and energy efficiency (EE), the CA-CNT and Borate-CNT catalysts induce 3.5% and 6% VE increase and 4% and 7% EE increase compared to without catalyst.

Fig. 7d ~ f represents the charge/discharge curves of Pristine VRFB, CA-CNT VRFB, and Borate-CNT VRFB. The IR-drop at 200 mA cm^{-2} of Pristine VRFB, CA-CNT VRFB, and Borate-CNT VRFB was 0.55, 0.48, and 0.45 V, respectively.

Fig. 7ã f shows that when the CA-CNT and Borate-CNT catalysts are used, the electron transfer between the vanadium ions and the electrodes occurs within lower overpotential and resistance, due to their ample number of active sites. Furthermore, a large amount of oxygen atoms in the Borate-CNT can produce the best VRFB performance.

It is also crucial to confirm the long term stability of the VRFB and the durability of the catalyst. To do that, the efficiencies of the VRFBs (Borate-CNT VRFB and Pristine VRFB) operated for 300 cycles in 200 mA cm^{-2} were measured (Fig. 8). According to Fig. 8, the VE of the Borate-CNT VRFB was consistently 74% even after 300 cycles, while the pristine VRFB gradually decreased from 74% to 60%. Moreover, CA-CNT VRFB showed 1.9% decrease of VE during in spite of only 50 cycles and lower current density (120 mA cm^{-2}) in previous work [26]. This can be explained since Borate-CNT can maintain its catalytic activity even in harsh acidic electrolyte conditions (2 M H_2SO_4) for a long

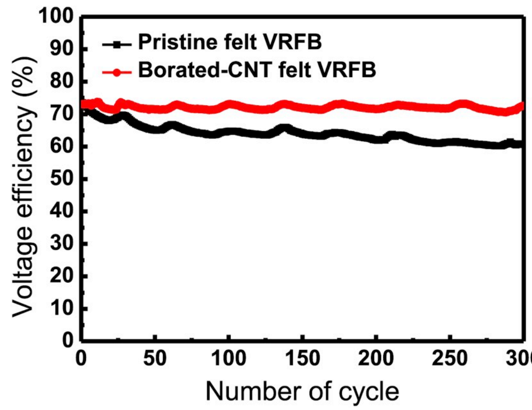


Fig. 8. The 300 cycle tests of Pristine VRFB and Borate-CNT VRFB. The cycles were performed in 200 mA cm^{-2} and their voltage efficiency was continuously recorded.

time. In addition, the Borate-CNT may play a critical role in protecting the carbon felt from chemical aging caused by the toxic acidic electrolyte. This proves that the Borate-CNT is an effective catalyst for enhancing the performance of VRFB, and an effective protector in suppressing the chemical aging of carbon felt from toxic acidic electrolytes. The damage and chemical aging of pristine felt were previously reported in other results and they were serious hurdles for the actual utilization of VRFB [28]. With the suggestion of the easily prepared Borate-CNT catalyst, we hope that the issues of VRFB can be addressed.

4. Conclusions

Borate-CNT was suggested as a catalyst to enhance the performance and long term durability of the VRFB. For this, the Borate-CNT catalyst was prepared by a simple, wet co-treatment process with NaOH and H_3BO_3. As a result, the amount of oxygen atoms in the catalyst significantly increased and the redox reactivity of the vanadium ions also improved dramatically by the use of the catalyst. According to characterization results, although catalysts fabricated with the single treatment of NaOH or H_3BO_3 also resulted in an increase in the amount of oxygen atoms and the catalytic activity, the co-treatment was the best way to

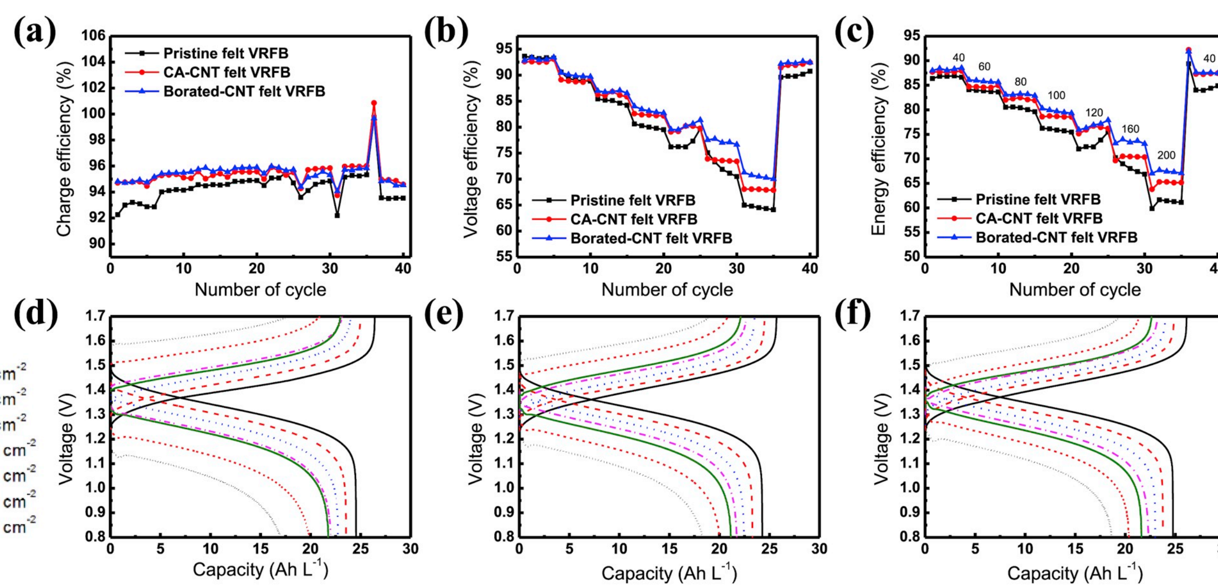


Fig. 7. (a) Charge efficiency, (b) voltage efficiency, and (c) energy efficiency of VRFBs using pristine felt, CA-CNT coated felt, and Borate-CNT coated felt for both electrodes. The operations were performed at different current densities of 40 → 60 → 80 → 100 → 120 → 160 → 200 → 40 mA cm^{-2} each for 5 cycles. (d) Charge/discharge curves of Pristine VRFB, (e) CA-CNT VRFB, and (f) Borate-CNT VRFB measured for the first cycle of 200 mA cm^{-2}.

improve the redox reactivity of the vanadium ions due to the increase of active sites by a formation of oxygen rich borate groups and a large difference in electronegativity between the boron and oxygen elements, which is an important factor to attract vanadium ions and facilitate their redox reactions.

According to the VRFB single cell test, it was demonstrated that the Borate-CNT VRFB showed better performance and durability results than the Pristine VRFB and CA-CNT VRFB. More specifically, the VE of the Borate-CNT VRFB was consistently maintained 74% even after 300 cycles, while the CA-CNT VRFB gradually decreased from 74% to 60%. This was because the Borate-CNT maintained its catalytic activity even in harsh acidic electrolyte conditions (2 M H_2SO_4) for a long time. In addition, it was verified that the Borate-CNT catalyst could protect the carbon felt from chemical aging attributed to the toxic acidic electrolyte. In conclusion, the Borate-CNT was an effective catalyst for improving the performance of VRFB and an effective protector in suppressing the chemical aging of carbon felt from a toxic acidic electrolyte.

Acknowledgement

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy(MOTIE) of the Republic of Korea (No. 20172420108550) and by the National Research Foundation of Korea (NRF) and the Ministry of the Ministry of Education (MOE) (No. 2019R1A2C1005776).

References

[1] M. Farrokhabadi, S. König, C.A. Cañizares, K. Bhattacharya, T. Leibfried, Battery energy storage system models for microgrid stability analysis and dynamic simulation, IEEE Trans. Power Syst. 33 (2018) 2301–2312, https://doi.org/10.1109/TPWRS.2017.2740163.
[2] F. Garcia-Torres, C. Bordons, M.A. Ridao, Optimal economic schedule for a network of microgrids with hybrid energy storage system using distributed model predictive control, IEEE Trans. Ind. Electron. 66 (2019) 1919–1929, https://doi.org/10.1109/TIE.2018.2826476.
[3] S.R. Shabanian, S. Edrisi, F.V. Khoram, Prediction and optimization of hydrogen yield and energy conversion efficiency in a non-catalytic filtration combustion reactor for jet A and butanol fuels, Korean J. Chem. Eng. 34 (2017) 2188–2197, https://doi.org/10.1007/s11814-017-0134-x.
[4] M. Masih-Tehrani, M.-R. Ha'iri-Yazdi, V. Esfahanian, A. Safaei, Optimum sizing and optimum energy management of a hybrid energy storage system for lithium battery life improvement, J. Power Sources 244 (2013) 2–10, https://doi.org/10.1016/J.JPOWSOUR.2013.04.154.
[5] J. Shen, S. Dusmez, A. Khaligh, Optimization of sizing and battery cycle life in battery/ultracapacitor hybrid energy storage systems for electric vehicle applications, IEEE Trans. Ind. Informatics 10 (2014) 2112–2121, https://doi.org/10.1109/TII.2014.2334233.
[6] L. Li, S. Kim, W. Wang, M. Vijayakumar, Z. Nie, B. Chen, J. Zhang, G. Xia, J. Hu, G. Graff, J. Liu, Z. Yang, A stable vanadium redox-flow battery with high energy density for large-scale energy storage, Adv. Energy Mater. 1 (2011) 394–400, https://doi.org/10.1002/aenm.201100008.
[7] C. Ponce de León, A. Frías-Ferrer, J. González-García, D.A. Szánto, F.C. Walsh, Redox flow cells for energy conversion, J. Power Sources 160 (2006) 716–732, https://doi.org/10.1016/J.JPOWSOUR.2006.02.095.
[8] E. Sum, M. Skyllas-Kazacos, A study of the V(II)/V(III) redox couple for redox flow cell applications, J. Power Sources 15 (1985) 179–190, https://doi.org/10.1016/0378-7753(85)80071-9.
[9] S. Zhong, M. Skyllas-Kazacos, Electrochemical behaviour of vanadium(V)/vanadium(IV) redox couple at graphite electrodes, J. Power Sources 39 (1992) 1–9, https://doi.org/10.1016/0378-7753(92)85001-Q.
[10] P. Han, H. Wang, Z. Liu, X. Chen, W. Ma, J. Yao, Y. Zhu, G. Cui, Graphene oxide nanoplatelets as excellent electrochemical active materials for VO^{2+}/VO_2^+ and V^{2+}/V^{3+} redox couples for a vanadium redox flow battery, Carbon N. Y. 49 (2011) 693–700, https://doi.org/10.1016/j.carbon.2010.10.022.
[11] W. Li, Z. Zhang, Y. Tang, H. Bian, T.-W. Ng, W. Zhang, C.-S. Lee, Graphene-nanowall-decorated carbon felt with excellent electrochemical activity toward VO_2^+/VO^{2+} couple for all vanadium redox flow battery, Adv. Sci. 3 (2016), https://doi.org/10.1002/advs.201500276, 1500276.
[12] W. Li, J. Liu, C. Yan, Multi-walled carbon nanotubes used as an electrode reaction catalyst for VO_2^+/VO^{2+} for a vanadium redox flow battery, Carbon N. Y. 49 (2011) 3463–3470, https://doi.org/10.1016/j.carbon.2011.04.045.
[13] G. Wei, C. Jia, J. Liu, C. Yan, Carbon felt supported carbon nanotubes catalysts composite electrode for vanadium redox flow battery application, J. Power Sources 220 (2012) 185–192, https://doi.org/10.1016/j.jpowsour.2012.07.081.
[14] L. Shi, S. Liu, Z. He, H. Yuan, J. Shen, Synthesis of boron and nitrogen co-doped carbon nanofiber as efficient metal-free electrocatalyst for the VO^{2+}/VO_2^+ Redox Reaction, Electrochim. Acta 178 (2015) 748–757, https://doi.org/10.1016/j.electacta.2015.08.026.
[15] C. Li, B. Xie, J. Chen, J. He, Z. He, Enhancement of nitrogen and sulfur co-doping on the electrocatalytic properties of carbon nanotubes for VO^{2+}/VO_2^+ redox reaction, RSC Adv. 7 (2017) 13184–13190, https://doi.org/10.1039/C6RA27734A.
[16] B. Kazeminasab, S. Rowshanzamir, H. Ghadamian, Nitrogen doped graphene/cobalt-based catalyst layers of a PEM fuel cell: performance evaluation and multi-objective optimization, Korean J. Chem. Eng. 34 (2017) 2978–2983, https://doi.org/10.1007/s11814-017-0202-2.
[17] X. Wu, H. Xu, P. Xu, Y. Shen, L. Lu, J. Shi, J. Fu, H. Zhao, Microwave-treated graphite felt as the positive electrode for all-vanadium redox flow battery, J. Power Sources 263 (2014) 104–109, https://doi.org/10.1016/J.JPOWSOUR.2014.04.035.
[18] D.M. Kabtamu, J.-Y. Chen, Y.-C. Chang, C.-H. Wang, Water-activated graphite felt as a high-performance electrode for vanadium redox flow batteries, J. Power Sources 341 (2017) 270–279, https://doi.org/10.1016/J.JPOWSOUR.2016.12.004.
[19] M.Z. Jelyani, S. Rashid-nadimi, S. Asghari, Treated carbon felt as electrode material in vanadium redox flow batteries : a study of the use of carbon nanotubes as electrocatalyst, J. Solid State Electrochem. (2016), https://doi.org/10.1007/s10008-016-3336-y.
[20] L. Dai, Y. Jiang, W. Meng, H. Zhou, L. Wang, Z. He, Improving the electrocatalytic performance of carbon nanotubes for VO^{2+}/VO_2^+ redox reaction by KOH activation, Appl. Surf. Sci. 401 (2017) 106–113, https://doi.org/10.1016/j.apsusc.2017.01.002.
[21] Z. Sun, Z. Wu, D. Liu, X. He, Microwave-assisted modification of activated carbon with cationic surfactants for enhancement of naphthalene adsorption, Korean J. Chem. Eng. 35 (2018) 557–566, https://doi.org/10.1007/s11814-017-0290-z.
[22] B. Sun, M. Skyllas-Kazacos, Modification of graphite electrode materials for vanadium redox flow battery application—I. Thermal treatment, Electrochim. Acta 37 (1992) 1253–1260, https://doi.org/10.1016/0013-4686(92)85064-R.
[23] X. Wu, H. Xu, P. Xu, Y. Shen, L. Lu, J. Shi, J. Fu, Microwave-treated graphite felt as the positive electrode for all- vanadium redox flow battery, J. Power Sources 263 (2014) 104–109, https://doi.org/10.1016/j.jpowsour.2014.04.035.
[24] J. Friedl, U. Stimming, Determining electron transfer kinetics at porous electrodes, Electrochim. Acta 227 (2017) 235–245, https://doi.org/10.1016/J.ELECTACTA.2017.01.010.
[25] M. Christwardana, J. Ji, Y. Chung, Y. Kwon, Highly sensitive glucose biosensor using new glucose oxidase based biocatalyst, Korean J. Chem. Eng. 34 (2017) 2916–2921, https://doi.org/10.1007/s11814-017-0224-9.
[26] M. Christwardana, Y. Chung, D.C. Tannia, Y. Kwon, Effects of the gold nanoparticles including different thiol functional groups on the performances of glucose-oxidase-based glucose sensing devices, Korean J. Chem. Eng. 35 (2018) 2421–2429, https://doi.org/10.1007/s11814-018-0163-0.
[27] C. Noh, S. Moon, Y. Chung, Y. Kwon, Chelating functional group attached to carbon nanotubes prepared for performance enhancement of vanadium redox flow battery, J. Mater. Chem. A. 5 (2017) 21334–21342, https://doi.org/10.1039/C7TA06672D.
[28] C. Noh, B.W. Kwon, Y. Chung, Y. Kwon, Effect of the redox reactivity of vanadium ions enhanced by phosphorylethanolamine based catalyst on the performance of vanadium redox flow battery, J. Power Sources 406 (2018) 26–34, https://doi.org/10.1016/J.JPOWSOUR.2018.10.042.
[29] C. Noh, C.S. Lee, W.S. Chi, Y. Chung, J.H. Kim, Y. Kwon, Vanadium redox flow battery using electrocatalyst decorated with nitrogen-doped carbon nanotubes derived from metal-organic frameworks, J. Electrochem. Soc. 165 (2018) A1388–A1399, https://doi.org/10.1149/2.0621807jes.
[30] L. Shao, G. Tobias, C.G. Salzmann, B. Ballesteros, S.Y. Hong, A. Crossley, B. G. Davis, M.L.H. Green, Removal of amorphous carbon for the efficient sidewall functionalisation of single-walled carbon nanotubes, Chem. Commun. 1 (2007) 5090–5092, https://doi.org/10.1039/b712614j.
[31] S. Musso, S. Porro, M. Vinante, L. Vanzetti, R. Ploeger, M. Giorcelli, B. Possetti, F. Trotta, C. Pederzolli, A. Tagliaferro, Modification of MWNTs obtained by thermal-CVD, Diam. Relat. Mater. 16 (2007) 1183–1187, https://doi.org/10.1016/J.DIAMOND.2006.11.087.
[32] Z. An, O.C. Compton, K.W. Putz, L.C. Brinson, S.T. Nguyen, Bio-inspired borate cross-linking in ultra-stiff graphene oxide thin films, Adv. Mater. 23 (2011) 3842–3846, https://doi.org/10.1002/adma.201101544.
[33] S. Liang, L. Liu, Q. Huang, K.L. Yam, Preparation of single or double-network chitosan/poly(vinyl alcohol) gel films through selectively cross-linking method, Carbohydr, Polymer 77 (2009) 718–724, https://doi.org/10.1016/J.CARBPOL.2009.02.007.
[34] X. Lu, W.L. Yim, B.H.R. Suryanto, C. Zhao, Electrocatalytic oxygen evolution at surface-oxidized multiwall carbon nanotubes, J. Am. Chem. Soc. 137 (2015) 2901–2907, https://doi.org/10.1021/ja509879r.
[35] J. Jin, X. Fu, Q. Liu, Y. Liu, Z. Wei, K. Niu, J. Zhang, Identifying the active site in nitrogen-doped graphene for the VO^{2+}/VO_2^+ redox reaction, ACS Nano 7 (2013) 4764–4773, https://doi.org/10.1021/nn3046709.
[36] E. Sum, M. Rychcik, M. Skyllas-kazacos, Investigation of the V(V)/V(IV) system for use in the positive half-cell of a redox battery, J. Power Sources 16 (1985) 85–95, https://doi.org/10.1016/0378-7753(85)80082-3.
[37] S.-K. Park, J. Shim, J.H. Yang, C.-S. Jin, B.S. Lee, Y.-S. Lee, K.-H. Shin, J.-D. Jeon, The influence of compressed carbon felt electrodes on the performance of a

vanadium redox flow battery, Electrochim. Acta 116 (2014) 447–452. https://doi.org/10.1016/j.electacta.2013.11.073.